Prathamesh PalveD20A 02

Blockchain Experiment No: 7

AIM: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

Blockchain is a decentralized and distributed digital ledger that records transactions securely across multiple systems. It ensures transparency, security, and immutability, meaning once data is recorded, it cannot be changed.

Ethereum is a popular blockchain platform that allows the execution of Smart Contracts. Smart contracts are self-executing programs written in Solidity that automatically perform actions when predefined conditions are met, eliminating the need for intermediaries.

MetaMask is a cryptocurrency wallet and browser extension used to interact with blockchain networks. It stores user accounts and private keys securely and allows users to send, receive, and sign transactions. It also acts as a bridge between Remix IDE and the blockchain network.

Testnets such as Sepolia and RSK Testnet are used for testing purposes. They provide free tokens, allowing developers to deploy and test smart contracts without using real cryptocurrency, reducing financial risk.

Remix IDE is a web-based development environment used to write, compile, deploy, and test smart contracts. It provides a user-friendly interface and supports direct integration with MetaMask for deploying contracts on blockchain networks.

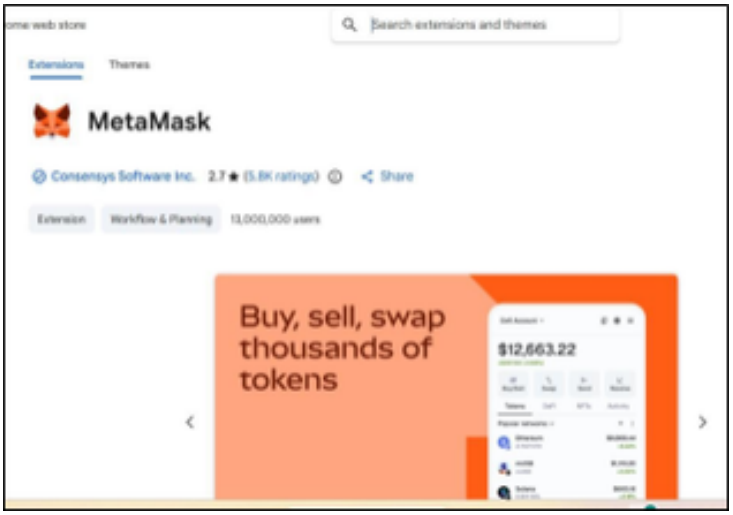
The smart contract deployment process includes writing the contract in Solidity, compiling it to remove errors, connecting MetaMask to Remix, and deploying the contract on a testnet by paying gas fees using test tokens.

A blockchain transaction refers to any interaction with the network, such as deploying a contract or calling its functions. Each transaction requires authentication through a digital signature and is permanently recorded on the blockchain. These transactions can be verified using blockchain explorers.

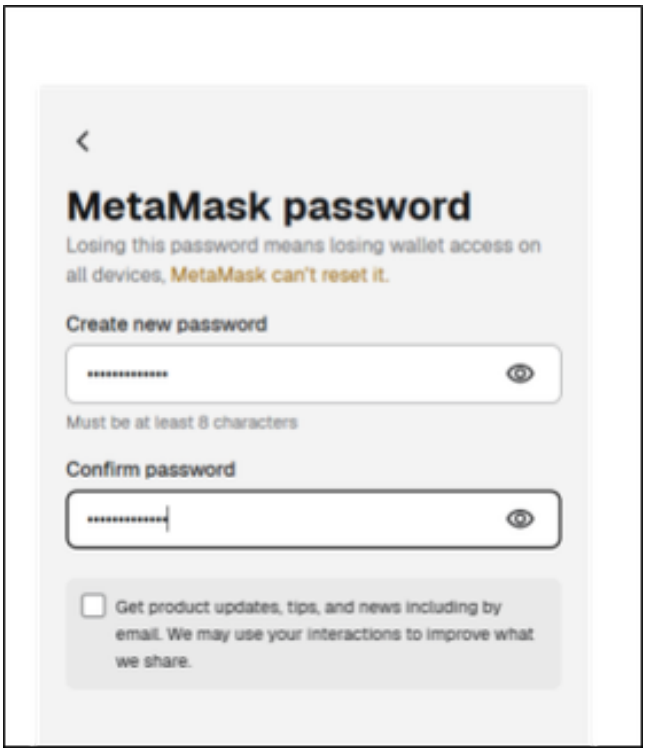
Implementation

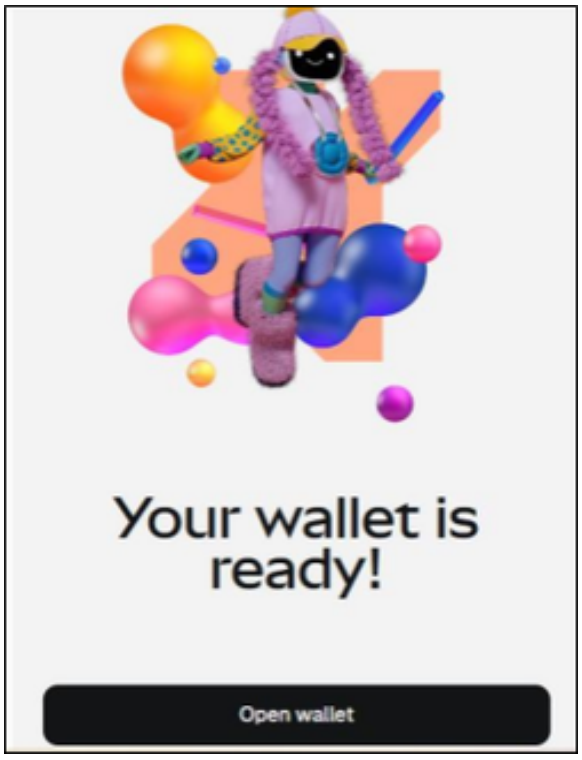
Step 1: Install MetaMask

Install MetaMask extension and create a wallet



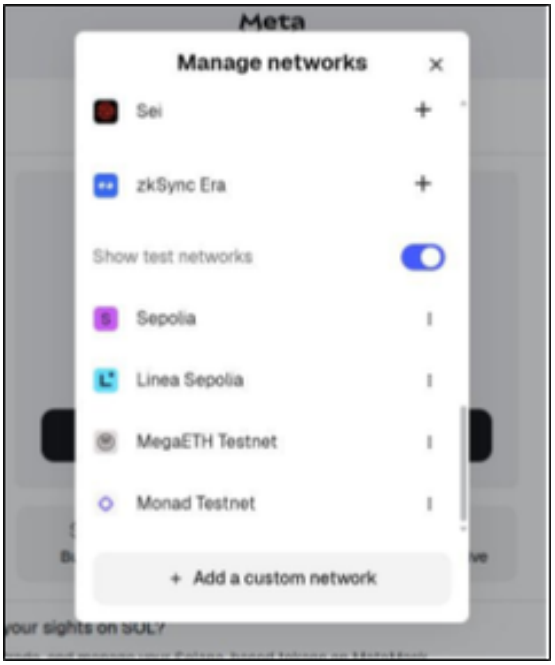
Meta
Mask
Create a new wallet
I have an existing wallet


MetaMask password
Losing this password means losing wallet access on all devices, MetaMask can't reset it.
Create new password
Must be at least 8 characters
Confirm password
Get product updates, tips, and news including by email. We may use your interactions to improve what we share.

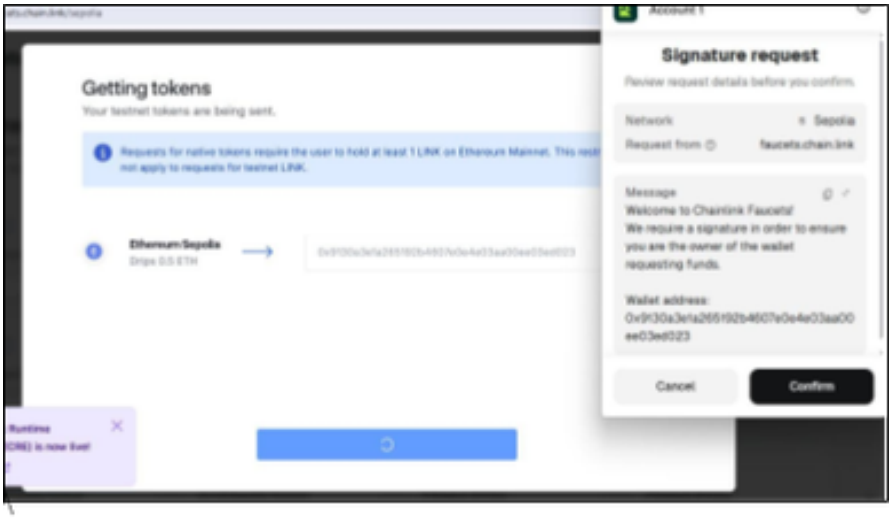


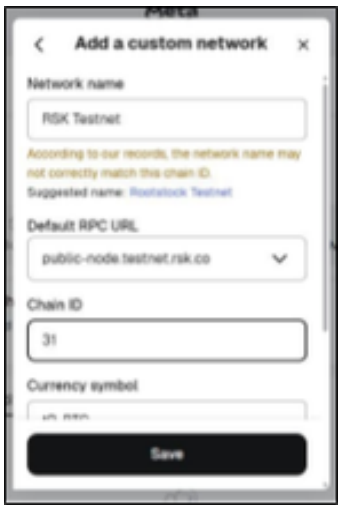
Step 2: Add Sepolia Testnet

Configure MetaMask to connect to Sepolia network.



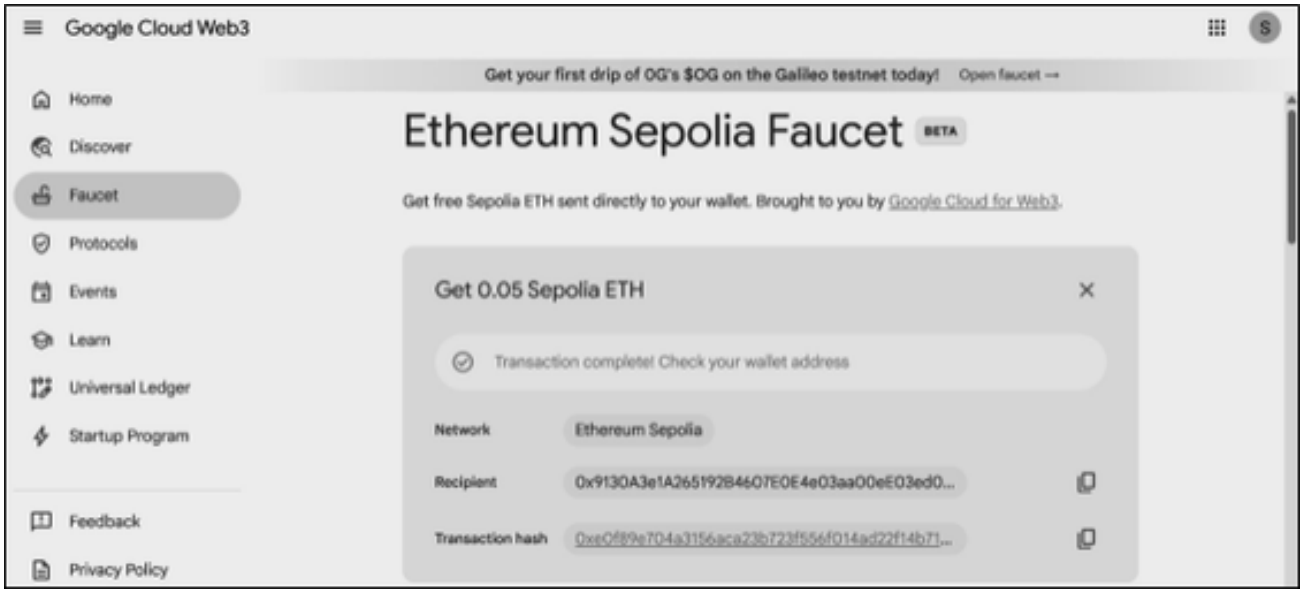
Edit network
Network name
Sepolia
Default RPC URL
Infura
sepolia.infura.io
Chain ID
11155111
Currency symbol
SepoliaETH
Save


Getting tokens
Your testnet tokens are being sent.
Ethereum/Sepolia
Signature request
Network
Sepolia
Request from
faucets.chain.link
Message
Welcome to Chainlink Faucets!
We require a signature in order to ensure you are the owner of the wallet requesting funds.
Wallet address:
Cancel
Confirm


Add a custom network
Network name
RSK Testnet
Default RPC URL
public-node.testnet.rsk.co
Chain ID
31
Currency symbol
Save

Step 3: Fund Wallet

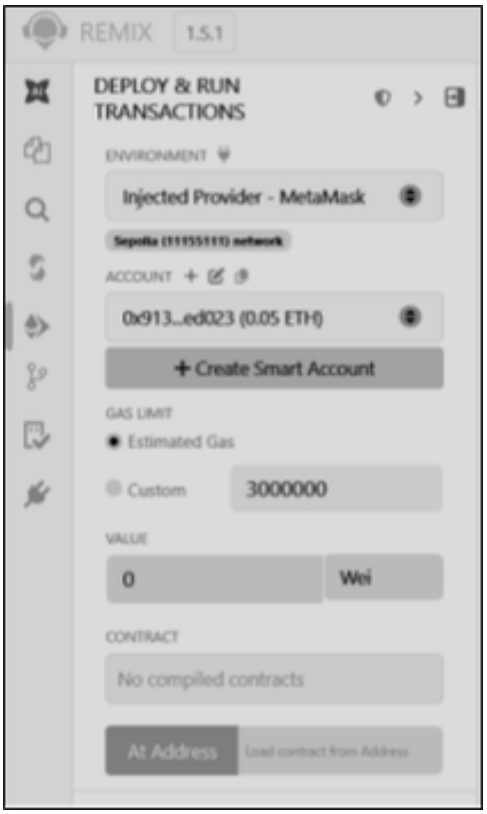
Use faucet to get test ETH



Step4: Connect the Sepolia Testnet / RSK Testnet to Remix IDE

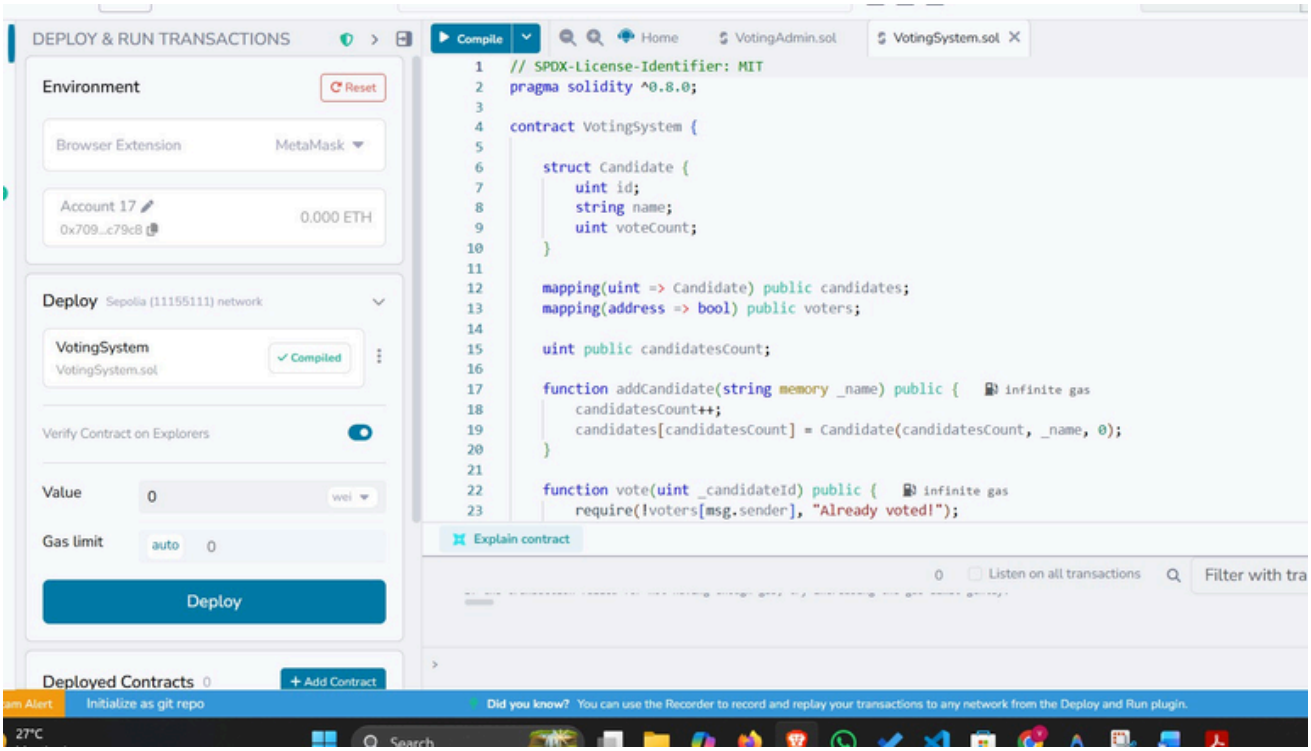
1. In the Remix IDE, select the Environment as Injected Web3.

Note : Injected Web3 connects Remix with the active account in Metamask.



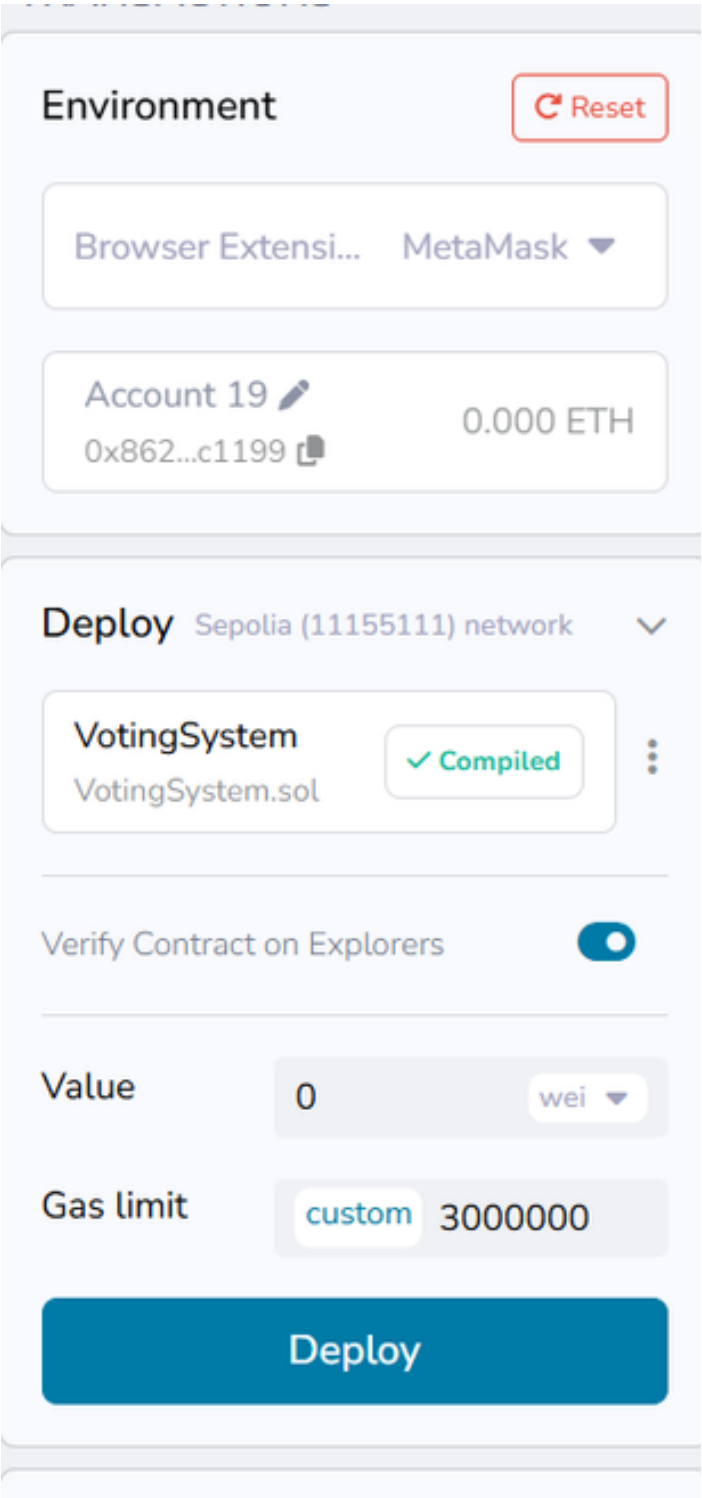
Compile the Contract

1. Open Remix
2. Paste your contract code
3. Go to Solidity Compiler (left panel)
4. Select version 0.8.x (match ^0.8.0)
5. Click Compile Votingsystem.sol



Connect MetaMask (Injected Provider)

1. Go to Deploy & Run Transactions tab
2. In Environment → select Injected Provider - MetaMask
3. MetaMask popup will open → click Connect Make sure:
   - Network = Sepolia Test Network
   - You have 0.05 Sepolia ETH

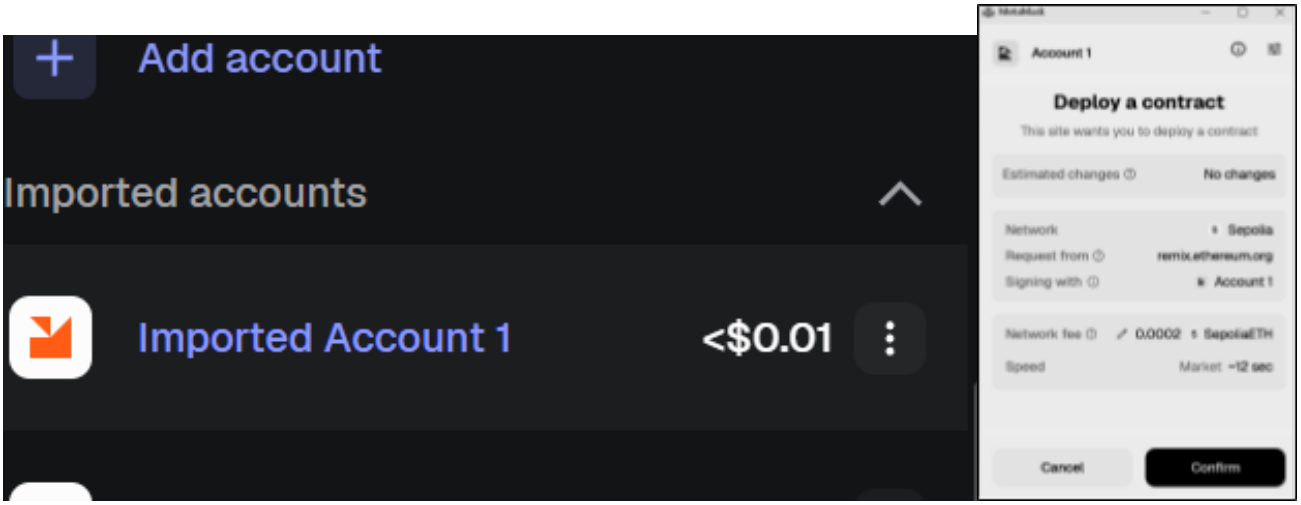


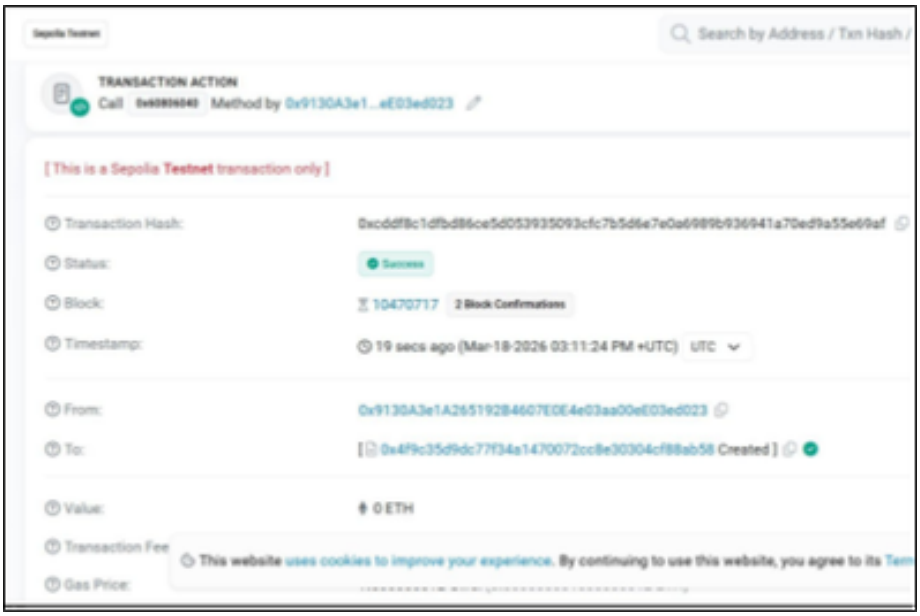
Deploy Contract

Your constructor needs a Voting address Steps:

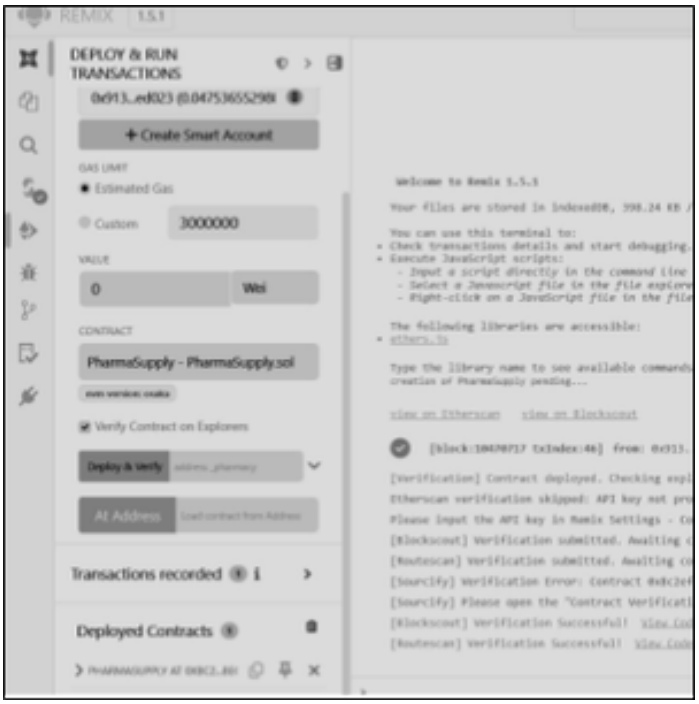
1. Copy your second MetaMask account address (voting)
   - MetaMask → Account 2 → Copy address
2. Paste that address in constructor field
3. Click Deploy
4. MetaMask will pop up → Confirm transaction Wait a few seconds…

After success → your contract appears under Deployed Contracts





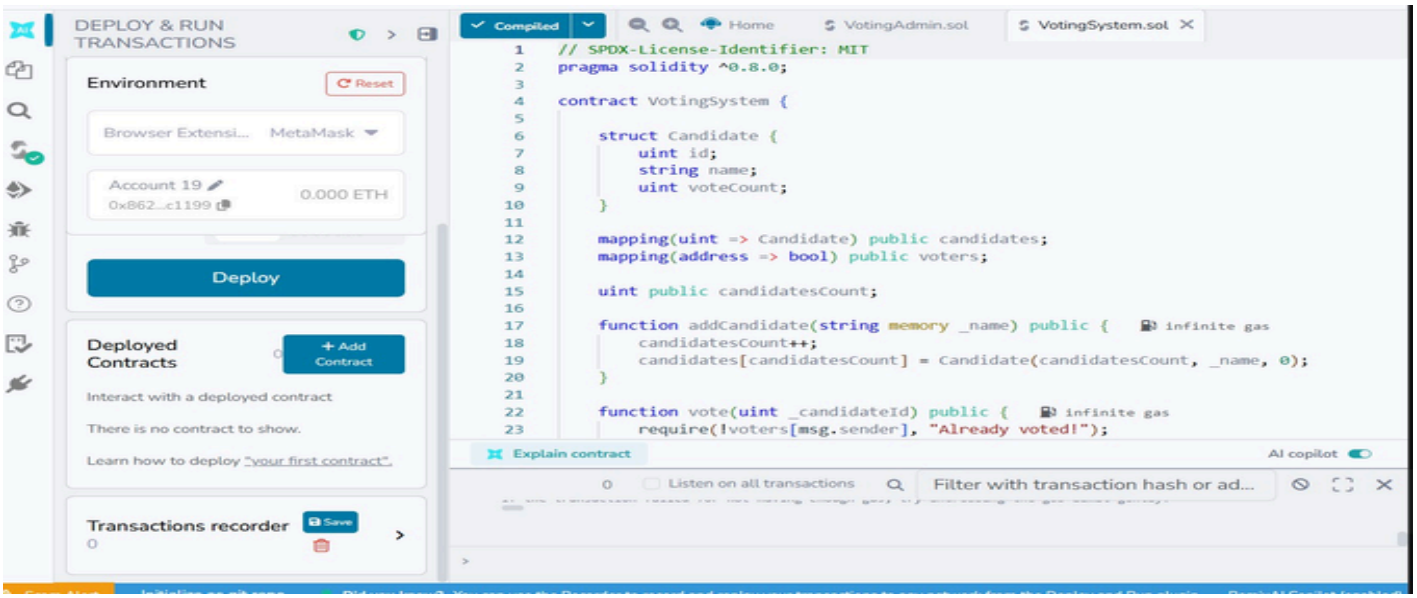
Contract deployment
Status
View on block explorer
Confirmed
Copy transaction ID
From
To
0x9130A..ed023
New contract
Transaction
Nonce 1
Amount -0 SepoliaETH
Gas Limit (Units) 828533
Gas Used (Units) 821549
Base fee (GWEI) 0.000000012
Priority fee (GWEI) 1.5
Total gas fee 0.001232 SepoliaETH
Max fee per gas 0.000000002 SepoliaETH
Total 0.00123172 SepoliaETH


REMIX 1.5.1
DEPLOY & RUN TRANSACTIONS
0x913..ed023 (0.047536552908
+ Create Smart Account
GAS LIMIT
Estimated Gas
Custom
3000000
VALUE
0
Wei
CONTRACT
PharmaSupply - PharmaSupply.sol
Verify Contract on Explorers
Deploy & Verify
At Address
Transactions recorded
Deployed Contracts
Welcome to Remix 1.5.1
You can use this terminal to:
The following libraries are accessible:

Interact with Contract

Step 1: Add Medicine (Manufacturer only)

- Make sure MetaMask is on Account 1 (manufacturer)
- In Remix:
  - Enter:
    - _candidate: "Alice"
- Click addCandidate Confirm in MetaMask



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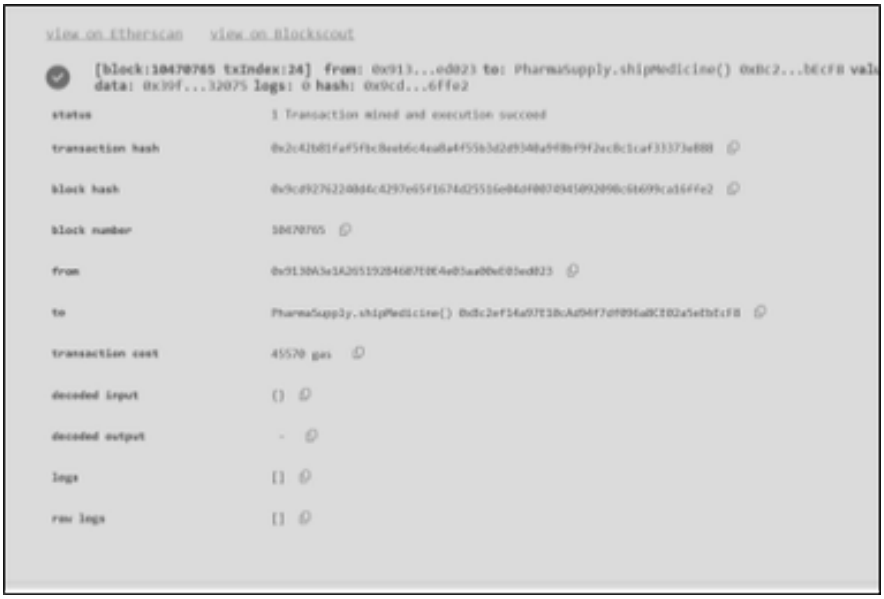
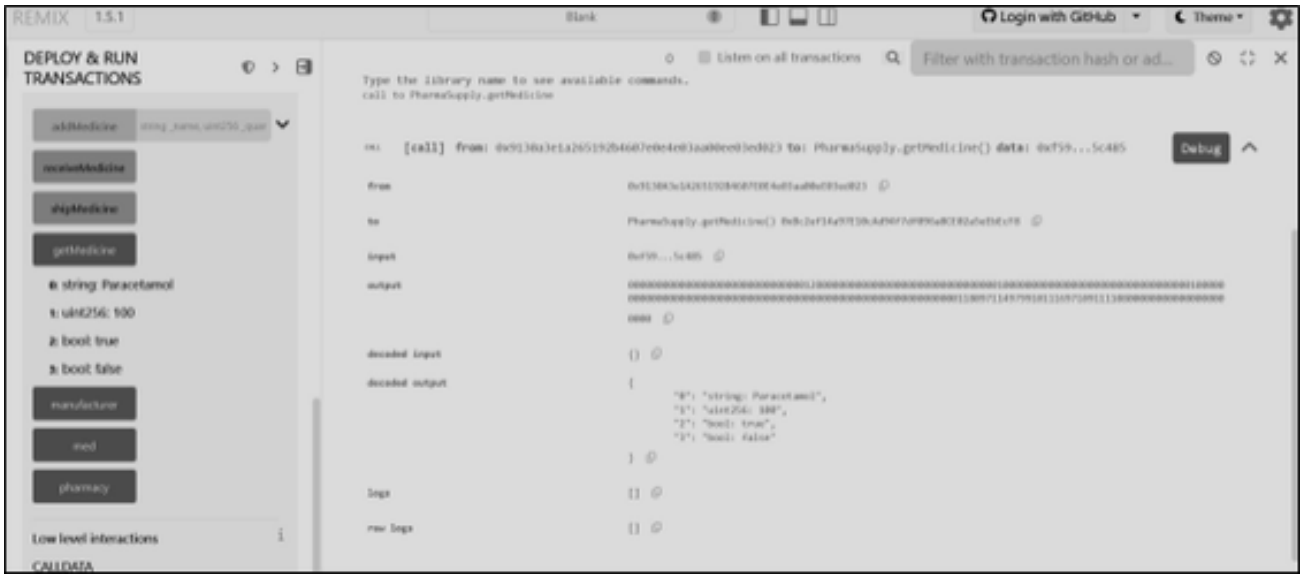
Step 2: Calculate Vote

- Still on Account 1
- Click calculateVote
- Confirm transaction





Conclusion

The experiment was successfully carried out by integrating the MetaMask wallet with Remix IDE and performing transactions on the Sepolia test network. The smart contract was compiled, deployed, and interacted with using MetaMask, which enabled secure transaction signing and real-time confirmation on the blockchain.

The use of a test network allowed safe execution without real financial risk, while also providing practical exposure to gas fees, transaction validation, and blockchain explorers. This experiment helped in understanding wallet integration, smart contract deployment, and the overall working of decentralized applications.

Thus, the objective of implementing MetaMask and performing blockchain transactions was successfully achieved.